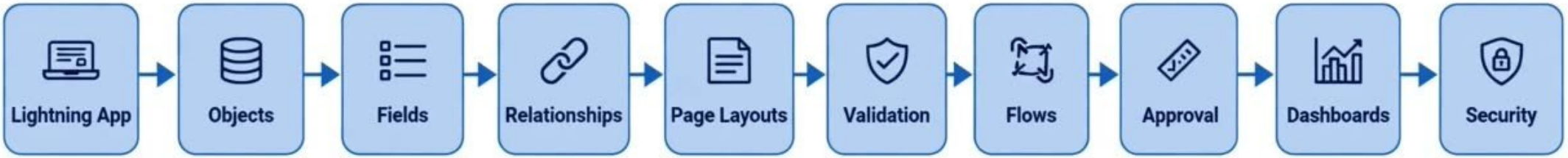
5. Project Development Phase - Code Layout & Readability

Section 5.2 Code Layout Principles for SECMS

Code Layout Principles (6 Principles)

1. **Object Naming Convention** — Use custom objects with suffix "_c". Examples: Student__c, Course__c, Enrollment__c, Instructor__c
2. **Field Naming** — Use clear label-based API names. Examples: Student_Name__c, Email__c, Phone__c
3. **Flow Naming** — Use descriptive action-based names. Examples: Assign_Instructor_Flow, Enrollment_Validation_Flow
4. **Modular Structure** — Separate flows for distinct responsibilities: validation, assignment, notification
5. **Validation Rules** — Implement one rule per logic with descriptive name to ensure clarity and maintainability
6. **Page Layout** — Organize into sections: Student Details | Academic Info | System Info

SECMS Development Workflow Diagram



5. Project Development Phase — Coding & Solution

SECMS — Student Enrollment & Course Management System

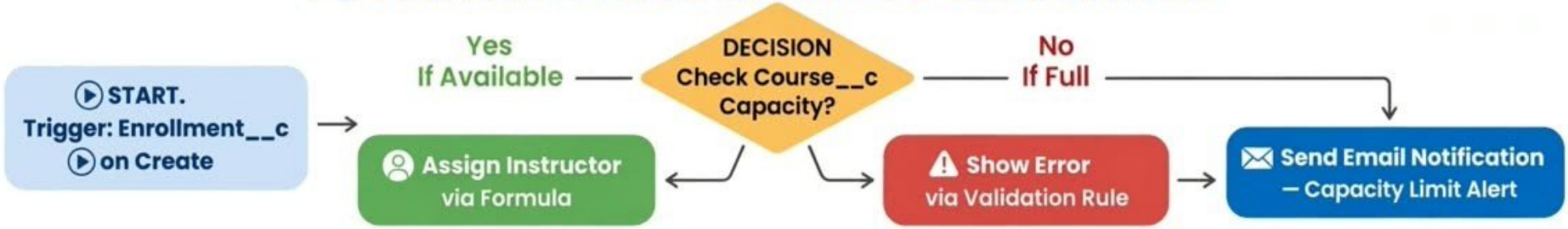
MILESTONE 1-14 FLOW:

- Salesforce Developer Account Creation
- Custom Object Creation (Student, Course, Enrollment, Instructor)
- Tabs Creation
- Lightning App Creation
- Fields Creation
- Create Relationships
- Page Layout Customization
- Validation Rules
- Flows (Automation)
- Approval Process
- Reports
- Dashboards
- Security Setup
- Conclusion

WHAT YOU'LL LEARN:

1. Real-time Salesforce project implementation
2. Designing custom objects and relationships
3. Creating page layouts & record types
4. Writing validation rules
5. Building automation with flows
6. Creating approval workflows
7. Developing reports & dashboards
8. Setting up security with profiles, permission sets, sharing
9. End-to-end application testing

EXAMPLE FLOW LOGIC DIAGRAM:



5. Project Development Phase - Functional Features

Overview of core functional modules, their features, associated objects/components, and user benefits for the SECMS platform

Functional Module	Feature Description	Objects / Components Involved	User Benefit
Student Management	Create, edit, view, delete student records with personal and academic details	Student__c object, Tabs, Page Layout	Quick admission processing ✓
Course Management	Create courses with capacity, duration, instructor assignment	Course__c object, Lookup to Instructor__c	Centralized course catalog ✓
Enrollment Processing	Student enrolls to course, status tracking Pending/Approved/Rejected, auto instructor assignment via flow	Enrollment__c Flow Assign_Instructor_Flow, Validation Rules	Automated enrollment with capacity check ✓
Instructor Management	Assign instructors to courses, track workload, view assigned students	Instructor__c Flow, Lookup	Balanced instructor workload ✓
Reports	Custom reports: Enrollments by Course, Students per Instructor, Course Capacity Utilization	Reports	Real-time insights for ✓ management
Dashboards	Visual charts: Total Enrollments, Course-wise Enrollment, Instructor Workload	Dashboards	Quick decision making ✓
Security	Role-based access: Admin, Enrollment Officer, Instructor, Student profiles, permission sets	Profiles, Permission Sets, Sharing Rules	Data protection ✓